

Data-Driven Sales, Inventory, and Profit Optimization in Agricultural Retail Business

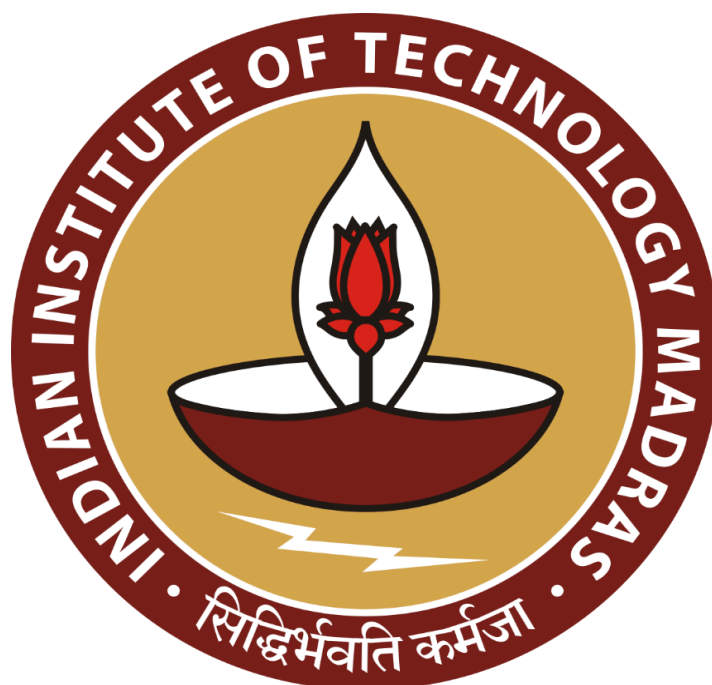
A Proposal report for the BDM capstone Project

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Declaration Statement

I am working on a Project titled "**Data-Driven Sales, Inventory and Profit Optimization in Agricultural Retail Business**". I extend my appreciation to **Krishi Khad Bhandar**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability. Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures. I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report. I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority. I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Chandan Kumar

Signature of Candidate: (Digital Signature)

Name: **Chandan Kumar**

Date: 11-10-2025

1. **Executive Summary**

Krishi khad bhandar is an agricultural retail business, located in Belhariya. The project focuses on family-run agricultural retail business and distribution of fertilizers, seeds and pesticides. The stores serve local farmers, dealers, retailer in nearby regions. Over the time business has accumulated valuable sales and product data that remain underutilized for decision making.

The business lacks insights into sales trends and cannot predict demand accurately across time, the business cannot identify which products drive the most revenue, There is no analysis of how pricing affects sales volume or total revenue.

The Business Data Management (BDM) Capstone Project plans to analyze the sales trend, optimizing inventory management by analyzing sales and purchase data, improving space utilization. By applying tools such as microsoft excel, inventory management software , statistical modeling, the analysis provides data driven recommendations for improving profitability, minimizing stockouts, and enhancing operational efficiency.

2. **Organization Background**

Krishi Khad Bhandar is a family-run agricultural retail business, located in Belhariya, Sarwan, Gaya, Bihar, 824201. The shop , owned by Mr. Ajesh kumar operates through business to business (B2B) and business to consumer (B2C) channels. It supplies essential farm inputs such as fertilizers, pesticides and seeds to customers across nearby rural regions. It provides high quality agricultural products sourced from reputable suppliers and manufacturers. The shop was established 20 years ago with an investment of 1 lakh. The company manages regular transactions of wholesalers, local retailers and individual farmers, maintaining inventory based on seasonal demand patterns. But the scale of operations has grown, managing sales data, inventory control, demand forecasting has become challenging.

3. Problem Statement

3.1 Sales trend analysis :- Unclear understanding of sales trends and seasonal demands fluctuation

3.2 Inventory Management Issues :- Difficulty in maintaining optimal stock levels, managing fluctuation in sales for perishable and seasonal products.

3.3 Pricing Optimization :- There is no analysis of how pricing affects sales volume and total revenue, examine correlation between unit price and quantity sold

3.4 Regional Sales Performance :- The business does not know which regions contribute most to overall revenue.

4. Background of the problem

4.1 Major Cause of the Problem

The primary cause of the problem is unclear understanding of sales trend analysis and seasonal demands fluctuation. Fluctuations in customer demands, seasonal trends, and market dynamics make it challenging to accurately forecast revenue and products that experience high seasonal and regional demand variation. It has difficulty in maintaining optimal stock levels.

4.2 Internal Problem

In the data driven world there are still family-run agricultural retailer shop rely on traditional methods for sales, inventory, customer demand. The business in focus deals with fertilizers, pesticides and seeds. Despite generating significant amount of sales data, the organization currently lacks a structured system to analyze and interpret this information effectively. Most records stored in manual ledgers limiting visibility in performance trends. As a result, the business struggles with challenges such as inaccurate demand forecasting, inefficient inventory control.

4.3 External Problem

The business operates in a highly seasonal and policy-sensitive environment, where external factors like weather conditions, government regulations, and

market competition significantly impact demand patterns, sales performance, and inventory management. These external challenges make it essential for the organization to adopt **data-driven tools** that improve forecasting accuracy and operational resilience.

5. Problem Solving Approach

5.1 Data collection and Understanding

The dataset used for this project consists of sales records from the agro-retail business, including details such as date, region, product category, unit price, quantity sold and total sales value. This data serves as the foundation for all analyses.

5.2 Data cleaning and preparation

Before analysis, the dataset undergoes cleaning and standardization to ensure accuracy. Excel will be used to check the data for missing values, inconsistencies, and any outliers. Using Excel filtering, sorting, and conditional formatting tools to quickly identify parts of the dataset that require quality checks. Summary statistics and pivot tools will provide an understanding of variable distributions.

5.3 Exploratory Data analysis

Using excel, sales data is visually explored to identify the patterns and trends. We conduct monthly and seasonal trends to explore which product sales is higher. Along with this analyses perform regional sales performance comparison and correlation between price and demand EDA helps uncover underlying business insights and performance gap.

5.4 Predictive and Analysis

Predictive analysis are done by Excel forecast function and python's time series models to forecast future sales demand. Forecasting helps the business to anticipate seasonal fluctuations, to plan which inventory demand high in particular period of time and improve stock management decisions to help business grow efficiently.

5.5 Actionable Insights and Recommendations

Based on the analytics, actionable recommendations will be created for Agri-retail business. Recommendations will include optimizing stock levels (e.g. ideal

quantity of stocks that meet customer demand), identifying profitable products (e.g. choose those inventory which profit more), improving sales strategies and pricing strategies (e.g. giving some discount for those who purchase maximum product).

Tools and Justification

Using Excel is the primary tool for our initial data cleaning, exploring, and visualizing is justified because it is easy to use and also fits definition of a business analytics tool. Excel based tracker will help visualize stock levels and turnover rates. We will also be using Python to perform more complex statistically analysis and advanced data manipulation. These tool enable efficient data processing, analysis, and presentation of actionable sights.

6. Expected Timeline

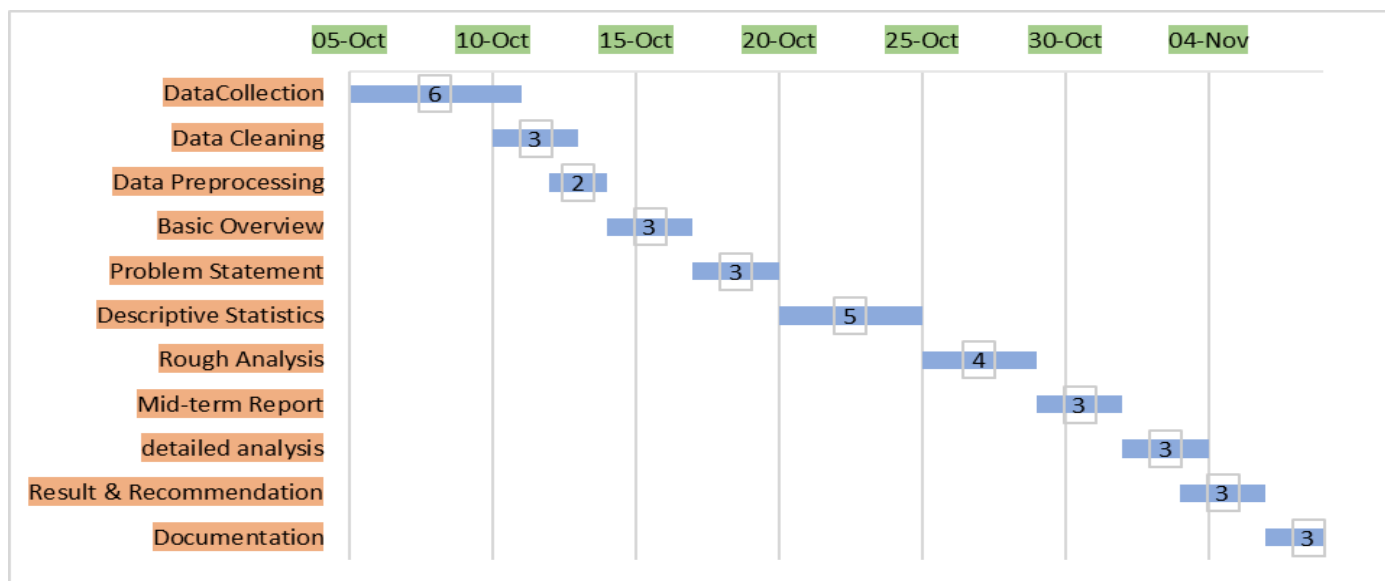


Figure 1: Gantt Chart of timeline

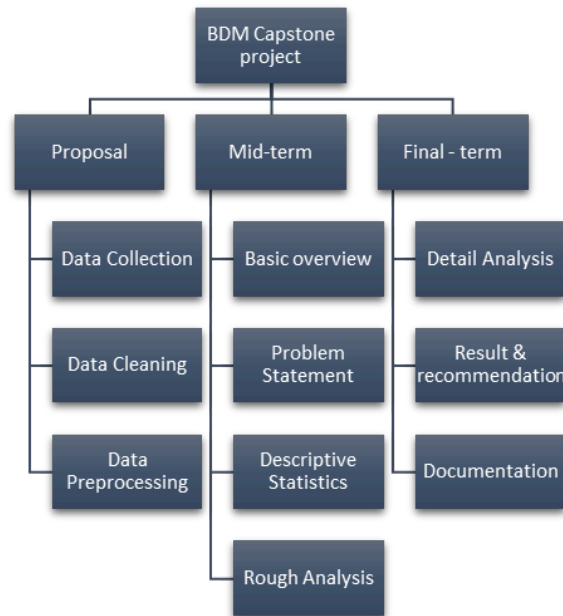


Figure 2: Work Breakdown Structure

7. Expected Outcome

- Identification of monthly, seasonal, and regional sales trends to guide business planning and marketing efforts.
- Ranking the high-performing and low-performing products and regions using ABC and comparative analysis.
- Accurate demand forecasts using historical data to support stock planning and prevent overstocking or shortages.
- Improve control over inventory levels, reducing carrying costs and ensuring product availability during peak demand Periods.
- Understanding of product level profitability and pricing sensitivity to enhance the revenue and margin optimization.